HG803 (HG-803) Temperature & Humidity Transmitter — Quick Guide

Overview

- **Outputs:** RS-485 Modbus RTU & 4–20 mA
- **Split probe option:** probe inside oxygen room, transmitter in control room
- **Accuracy:** ±0.3 °C, ±3 %RH
- **Power:** 10–30 V DC
- **Range:** −40 °C to +80 °C, 0–100 %RH

Installation (Oxygen Generator Room)

- Use **split-probe model** → only probe in oxygen room.
- Run probe cable through **sealed conduit**.
- Place transmitter in control room.
- Connect RS485 A/B lines to PLC/SCADA or 4–20 mA to analog input.

Wiring Diagram

```
[Probe in Oxygen Room] —— cable —— [HG803 Transmitter Box in Control Room]
                                      |-- Power 12–24 VDC
                                      |-- RS485 A/B → PLC/SCADA
                                      |-- 4–20 mA → Analog Input
```

Modbus Example

- **ID:** configurable (default 1)
- **Baud:** 9600
- **Registers (typical):**
- 40001 → Temperature (°C)
- 40002 → Humidity (%RH)

Price & Warranty

- **Price:** $30–$90 basic, $70–$200 better builds
- **Warranty:** 6–12 months (vendor dependent)
- **Lifespan:** 5–10 years (probe replaceable)

Maintenance

- **Monthly:** check vs reference meter

- **6 months:** calibration check
- **Annually:** clean probe filter
- **3–5 years:** replace probe

Notes

- Not intrinsically safe — only probe should enter oxygen room.
- Confirm local safety regulations before installation.